Type MS · Quadratic inequalities

Answer 1 of 10 · Question 1 part b · 4 marks

2023 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Use of a formula:

Generally, the method mark is gained by **either**

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Answer 1 of 10 · Question 1 part b · 4 marks

2023 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."**Exact answers:**

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

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Answer 1 of 10 · Question 1 part b · 4 marks

2023 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Paper 2		
Question number	Scheme	Marks
1 a	$9x < 6 \Rightarrow x < \ldots$	M1
	$x < \frac{2}{3}$	A1 (2)
b	$(3x+1)(x-3) < 0$	M1
	$x = -\frac{1}{3} \quad x = 3$	A1
	$-\frac{1}{3} < x < 3$	M1 A1 (4)
c	$-\frac{1}{3} < x < \frac{2}{3}$	B1ft (1)
		Total 7 marks

Type MS · Quadratic inequalities

Answer 2 of 10 · Question 2 part b · 4 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

2 (a)	$f(x)=2\left(x^2-6x\right)+5$ OR $f(x)=2\left(x^2-6x+\frac{5}{2}\right)$	M1
	$f(x)=2\left(\left[x-3\right]^2-k\right)+5$ OR $f(x)=2\left(\left[x-3\right]^2-k+\frac{5}{2}\right)$	M1
	$f(x)=2\left[x-3\right]^2-13$	A1
	$a=2,\ b=-3,\ c=-13$ Allow values embedded in f (x)	[3]
ALT – Equating coefficients		
	$ax^2+2abx+\left(ab^2+c\right)\equiv 2x^2-12x+5\Rightarrow a=2$	[M1
	$2ab=-12\Rightarrow 2\times 2\times b=-12\Rightarrow b=-3$	M1
	$ab^2+c=5\Rightarrow 2\times(-3)^2+c=5\Rightarrow c=5-18=-13$	A1]
(b)	$2\left[x-'3'\right]^2-'13'>37\Rightarrow 2\left[x-'3'\right]^2=50\Rightarrow\left[x-3\right]^2=25\Rightarrow x='3'\pm\ldots$	M1
	Critical values $x=-2,8$	A1
	$\{x<-2\}\cup\{x>8\}$	M1A1 [4]
ALT - solving inequality without use of completed square form		
	$2x^2-12x-32>0\ \Rightarrow\ x^2-6x-16>0\ \Rightarrow\ (x+2)(x-8)>0$	[M1
	Critical values $x=-2,8$	A1
	$\{x<-2\}\cup\{x>8\}$	M1A1]
		Total 7 marks

Type MS · Quadratic inequalities

Answer 2 of 10 · Question 2 part b · 4 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

		$ab^2+c=5 \Rightarrow 2\times(-3)^2+c=5 \Rightarrow c=5-18=-13$
	A1	For correct values of a, b and c. $a=2,\ b=-3,\ c=-13$ Allow embedded answers i.e. $\mathrm{f}(x)=2[x-3]^2-13$
(b)	**M1**	Sets f(x) > 37 and uses their result from part (a) [provided it is in the form $\mathrm{f}(x)=2(x\pm P)^2\pm Q$] and attempts to find **two** critical values $2['x-'3']^2-'13'>37 \Rightarrow 2['x-'3']^2=50 \Rightarrow [x-3]^2=25 \Rightarrow x='3'\pm\ldots$ Condone use of = rather than >
	A1	For both critical values of $x=-2,\ 8$
	M1	For choosing the outside region for their cv's [provided there are two values] $\{x<'-2'\}\cup\{x>'8'\}$ or any other correct notation. Condone 'AND' for this mark
	A1	For the correct region with correct values $\{x<-2\}\cup\{x>8\}$ or any other correct notation. Must **not** use 'AND' for this mark.
Alt – solving inequality without use of completed square form		
	M1	Sets f(x) > 37 and attempts to find **two** critical values. See general guidance on what constitutes an attempt to solve a quadratic. Condone use of = rather than >
	A1	For both critical values of $x=-2,\ 8$
	M1	For choosing the outside region for their cv's [provided there are two values] $\{x<'-2'\}\cup\{x>'8'\}$ or any other correct notation. Condone 'AND' for this mark
	A1	For the correct region with correct values $\{x<-2\}\cup\{x>8\}$ or any other correct notation. Must **not** use 'AND' for this mark.

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Type MS · Quadratic inequalities

Answer 2 of 10 · Question 2 part b · 4 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Quadratic inequalities

Answer 3 of 10 · Question 1 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

1. Factorisation:

$(x^2+bx+c)=(x+p)(x+q)$, where $|pq|=|c|$ leading to $x=....$

$(ax^2+bx+c)=(mx+p)(nx+q)$ where $|pq|=|c|$ and $|mn|=|a|$ leading to $x=....$

2. Formula:

Attempt to use the **correct** formula (shown explicitly or implied by working) with values for *a*, *b* and *c*, leading to $x=....$

3. Completing the square:

$x^2+bx+c=0$: $\left(x\pm\frac{b}{2}\right)^2\pm q\pm c=0,\quad q\neq 0$ leading to $x=....$

Method marks for differentiation and integration:

1. Differentiation

Power of at least one term decreased by 1. $\left(x^n\rightarrow x^{n-1}\right)$

2. Integration:

Power of at least one term increased by 1. $\left(x^n\rightarrow x^{n+1}\right)$

Type MS · Quadratic inequalities

Answer 3 of 10 · Question 1 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

Use of a formula:

Generally, the method mark is gained by **either**

quoting a correct formula and attempting to use it, even if there are mistakes in the substitution of values

or, where the formula is not quoted, the method mark can be gained by implication from the substitution of correct values and then proceeding to a solution.

Answers without working:

The rubric states "Without sufficient working, correct answers may be awarded no marks".

General policy is that if it could be done "in your head" detailed working would not be required. (Mark schemes may override this eg in a case of "prove or show...."

Type MS · Quadratic inequalities

Answer 3 of 10 · Question 1 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

Exact answers:

When a question demands an exact answer, all the working must also be exact. Once a candidate loses exactness by resorting to decimals the exactness cannot be regained.

Rounding answers (where accuracy is specified in the question)

Penalise only once per question for failing to round as instructed - ie giving more digits in the answers. Answers with fewer digits are automatically incorrect, but the isw rule may allow the mark to be awarded before the final answer is given.

Type MS · Quadratic inequalities

Answer 3 of 10 · Question 1 part b · 3 marks

2022 JAN · Paper 02 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks
1 (a)	$x > 3$	B1 (1)
(b)	$(2x-1)(x-5)$	M1
	Critical values are $x = \frac{1}{2}$ and $x = 5$	A1
	$\frac{1}{2} < x < 5$	B1ft (3)
(c)	$3 < x < 5$	B1ft (1)
		Total 5 marks

Type MS · Quadratic inequalities

Answer 4 of 10 · Question 2 part b · 4 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

2(a)	$3x-8<5x+3 \Rightarrow x>-\frac{11}{2}$	B1 [1]
(b)	$4x^2-7x+1>6-2x^2 \Rightarrow$ $6x^2-7x-5>0 \Rightarrow (3x-5)(2x+1)>0$ cv's are $x=\frac{5}{3},\ -\frac{1}{2} \Rightarrow x<-\frac{1}{2},\ x>\frac{5}{3}$	M1M1 M1A1 [4]
(c)	$-\frac{11}{2}<x<-\frac{1}{2},\ x>\frac{5}{3}$	B1 ft [1]
	Total	**6 marks**

Question	Notes	Marks
2(a)	$3x-8<5x+3$ $\Rightarrow x>-\frac{11}{2}$	B1 [1]
(b)	For rearranging to form a 3TQ $4x^2-7x+1>6-2x^2 \Rightarrow 6x^2-7x-5>0$	M1
	For method to solve their 3TQ. See general guidance for what constitutes an attempt to solve. $6x^2-7x-5>0 \Rightarrow (3x-5)(2x+1)>0$	M1
	For writing down their critical values from their factorisation and the correct inequality for their cv's [outside region]. cv's are $x=\frac{5}{3},\ -\frac{1}{2}$ $x<-\frac{1}{2},\ x>\frac{5}{3}$	M1
	For the correct inequalities only $x<-\frac{1}{2},\ x>\frac{5}{3}$ A0 if rejects one region or if inequalities are incorrectly combined.	A1 [4]
(c)	For the correct inequality only $-\frac{11}{2}<x<-\frac{1}{2},\ x>\frac{5}{3}$ For follow through (a) must be a linear inequality and (b) must be of the form $x<a, x>b$ with $b>a$.	B1 ft [1]
	Total	**6 marks**

Type MS · Quadratic inequalities

Answer 4 of 10 · Question 2 part b · 4 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 1 of 15 · find part b in this mark scheme extract

Question	Scheme	Marks
4a)	$\theta = \dfrac{21-r}{r}$ $A = \dfrac{r^2}{2}\theta \Rightarrow A = \dfrac{r^2}{2}\left(\dfrac{21-r}{r}\right) \Rightarrow A = \dfrac{r}{2}(21-r)*$ **ALT** OR $\theta r = 21-r$ or $l = 21-r$ $A = \dfrac{r}{2}r\theta$ or $A = \dfrac{1}{2}rl \Rightarrow A = \dfrac{r}{2}(21-r)*$	B1 M1A1 cso [3] [B1 M1A1]
(b)	$27 \leqslant \dfrac{r}{2}(21-r) \Rightarrow 54 \leqslant 21r - r^2 \Rightarrow r^2 - 21r + 54 \leqslant 0$ Finds c.v's $r^2 - 21r + 54 = 0 \Rightarrow (r-18)(r-3) = 0 \Rightarrow r = 18, 3$ $3 \leqslant r \leqslant 18$	B1 M1 M1A1 [4]
(c)	$\theta = \dfrac{21-r}{r} \Rightarrow \theta = \dfrac{21-18}{18} = \dfrac{1}{6} \quad \theta = \dfrac{21-3}{3} = 6$ $\Rightarrow \dfrac{1}{6} \leqslant \theta \leqslant 6$	B1ft B1 [2]
		Total 9 marks

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 2 of 15 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	**Mark parts (i) and (ii) together.** **If they find *a* first score M1M1M1-**	
	M1	For using the correct formula to write down: $4k+1=a+(6-1)d$
	M1	For using the correct formula to write down: $36k+1=\frac{10}{2}(2a+9d)$
(i)	M1	For an explicit, complete method to solve their two simultaneous equations to find a value for d in terms of k, and a value for a Allow **one** processing error. **Note: This is a show question**
(ii)	A1	For the correct expression for d
	A1 cso	For finding $a=-8$ With no errors, this is a given value. Award this if they find only a but not d
(b)	M1	Uses a correct nth term with their expression for d to form a linear equation **and** attempts to solve it.
	A1 cso	For $k=4$ With no errors, this is a given value.
(c)	M1	For setting up the equation as required using the given a and their d No simplification is required for this mark.
	dM1	For forming a 3TQ using the given a and their d Allow one error: $5n^2-71n-K=0$ Or $5n^2-Ln-580=0$ Or $Mn^2-71n-580=0$ Where K, L and M are constants. Note: this mark is dependent on the previous mark being scored.
	M1	For solving their 3TQ using any valid, correct method If they use a calculator and the 3TQ is correct together with the correct final value, award this mark. If the 3TQ is NOT correct, they must show us a correct valid method for solving the 3TQ for the award of this mark.
	A1	For $n=20$ with no other value.

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 3 of 15 · find part b in this mark scheme extract

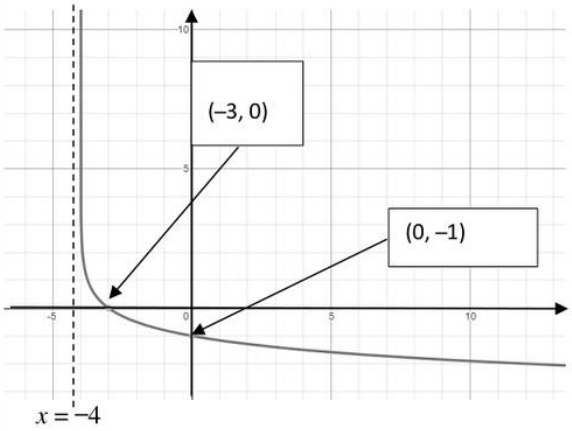
Question	Scheme	Marks
6(a)	$t=0 \Rightarrow s=e^{2\times 0}\sin(3\times 0)+2=2$ $t=\frac{\pi}{6} \Rightarrow s=e^{2\times\frac{\pi}{6}}\sin\left(3\times\frac{\pi}{6}\right)+2=e^{\frac{\pi}{3}}+2$ $AB=e^{\frac{\pi}{3}}+2-2=e^{\frac{\pi}{3}}$ (m)	M1 A1 [2]
(b)	$v=2e^{2t}\sin 3t+3e^{2t}\cos 3t$ When $t=\frac{\pi}{3}$ $v=2e^{\frac{2\pi}{3}}\sin\left(\frac{\pi}{3}\times 3\right)+3e^{\frac{2\pi}{3}}\cos\left(\frac{\pi}{3}\times 3\right)=-3e^{\frac{2\pi}{3}}$ (m/s)	M1A1A1 B1ft [4]
		Total 6 marks

Part	Mark	Notes
(a)	M1	For an attempt to find the displacements of A **and** B as simplified values. At least one must be correct and simplified for this mark and there must be an attempt at the other.
	A1	For finding the distance AB
(b)	M1	For an attempt at product rule. • The formula must be correct. i.e., $[v]=ke^{2t}\times\sin 3t+e^{2t}\times l\cos 3t$ • There must be an acceptable attempt to differentiate both e^{2t} and $\sin 3t$ $e^{2t}\to ke^{2t}$ and $\sin 3t\to l\cos 3t$ $k,l\neq 0$ and integers • The terms must be added.
	A1	One term fully correct.
	A1	Both terms fully correct.
	B1ft	For using $t=\frac{\pi}{3}$ in their v to find a value for v The minimally acceptable expression for v must be a **changed** expression from the given s in terms of e^{2x}, $\sin 3x$ and $\cos 3x$ Note: This is a ft mark. Please check their substitution and the final result. Please isw a value of $-24.36...$ if seen **together** with the exact value.

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

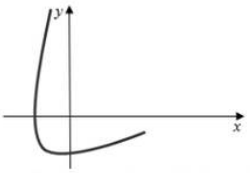
2024 JUN · Paper 01 (regional R) · extract 4 of 15 · find part b in this mark scheme extract

Question	Scheme	Marks
7(a)		B1 Shape and position B1 Intersection with x-axis B1 Intersection with y-axis B1 Asymptote [4]
(b)	$\log_{(x+4)} 256 - \log_4(x+4) = 0 \Rightarrow \dfrac{\log_4 256}{\log_4(x+4)} - \log_4(x+4) = 0$ $\Rightarrow 4 - (\log_4(x+4))^2 = 0 \Rightarrow (\log_4(x+4))^2 = 4$ $\Rightarrow \log_4(x+4) = \pm 2$ $x+4 = 4^2 \Rightarrow x = 12$ $x+4 = 4^{-2} \Rightarrow x = -\dfrac{63}{16}$ or -3.9375	M1 M1 M1 A1 A1 [5]
		Total 9 marks

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 5 of 15 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct shape in the correct position. Ignore intersections and the asymptote for this mark. Award as long as the curve is the correct way around in quadrants 2, 3 and 4. The ends must not turn back in themselves. Do Not accept for example:  If there are two or more lines, and candidates have not made clear which ONE they want marked– award B0
	B1	For a line/curve **passing** through $(-3, 0)$ or accept $x = -3$ Do not accept stopping at the axis.
	B1	For a line/curve passing through $(0, -1)$ or accept $y = -1$ Do not accept stopping at the axis.
	B1	For the correct **equation** only of the asymptote. If they also give a horizontal asymptote then do not isw – it is B0
(b)	**Method 1 - Works in log base 4**	
	M1	For changing the base of the log to base 4
	M1	For forming a quadratic in terms of $(x + 4)$ Accept a substitution for the log
	M1	For taking the square root **and** undoing the log. Accept just the positive root for this mark.
	A1	For either $x = 12$ or $-\frac{63}{16}$
	A1	For both $x = 12$ and $-\frac{63}{16}$
	Method 2 – Works in log base $(x + 4)$	
	M1	For changing the base of the log to base $(x + 4)$ $\log_{(x+4)} 256 - \frac{\log_{(x+4)}(x+4)}{\log_{(x+4)} 4} = 0 \Rightarrow$
	M1	Changes $\log_{(x+4)}(x+4) = 1$ and $\log_{(x+4)} 256 = 4\log_{(x+4)} 4$ And forming a quadratic equation with log base (x + 4) $\left(\log_{(x+4)} 4\right)^2 = \frac{1}{4}$ Accept a substitution for the log
	M1	Takes the square root (accept just the positive root) **and** undoes the log $\log_{(x+4)} 4 = \pm\frac{1}{2} \Rightarrow 4 = (x+4)^{\frac{1}{2}}$ and $4 = (x+4)^{-\frac{1}{2}}$
	A1	For either $x = 12$ or $-\frac{63}{16}$
	A1	For both $x = 12$ and $-\frac{63}{16}$

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

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	Method 3 works in an unspecified base. If it is just log, assume base 10	
	M1	For changing the base $\frac{\log 256}{\log(x+4)} - \frac{\log(x+4)}{\log 4} = 0$
	M1	Forms a QE in terms of $(x+4)$ $\quad 4(\log 4)^2 = (\log(x+4))^2$ oe
	M1	Takes square root (accept just the positive root) and undoes the log $\pm 2\log 4 = \log(x+4) \Rightarrow \log 4 = \pm\frac{1}{2}\log(x+4)$ $\Rightarrow \log 4 = \log(x+4)^{\pm\frac{1}{2}} \Rightarrow 4 = (x+4)^{\pm\frac{1}{2}}$ $\Rightarrow 16 = x+4, \quad \frac{1}{16} = x+4$
	A1	For either $x = 12$ or $-\frac{63}{16}$
	,A1	For both correct values $x = 12$ and $-\frac{63}{16}$

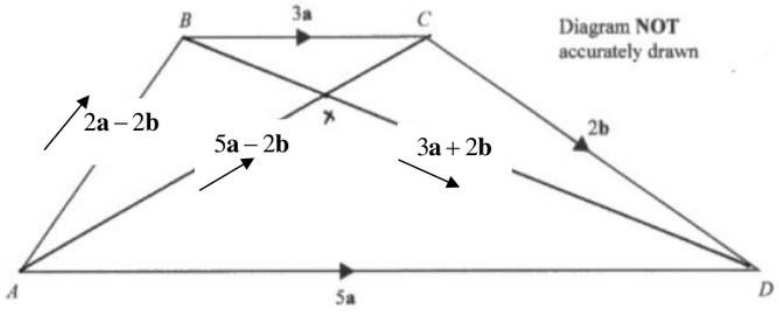
Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 7 of 15 · find part b in this mark scheme extract

Question	Scheme	Marks
8(a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DC} + \overrightarrow{CB} = 5\mathbf{a} - 2\mathbf{b} - 3\mathbf{a} = 2\mathbf{a} - 2\mathbf{b}$	B1 [1]
(b)	$\overrightarrow{BX} = k\overrightarrow{BD} = k(3\mathbf{a} + 2\mathbf{b})$	M1
	$\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$	M1
	$\Rightarrow k(3\mathbf{a} + 2\mathbf{b}) = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow 3k\mathbf{a} + 2k\mathbf{b} = (5\lambda - 2)\mathbf{a} + (2 - 2\lambda)\mathbf{b}$ $\Rightarrow 3k = 5\lambda - 2 \quad 2k = 2 - 2\lambda$	M1
	$\Rightarrow k = \frac{3}{8} \quad \left[\lambda = \frac{5}{8}\right]$	M1A1 [5]
(c)	$\Delta CXD = \frac{5}{8}\Delta BCD$	M1
	$\Delta BCD = \frac{3}{8}ABCD$	M1
	$\frac{\Delta CXD}{ABCD} = \frac{\frac{5}{8}}{\frac{3}{8}} = \frac{15}{64}$	M1
	Ratio of area of triangle CXD : Trapezium $ABCD$ = 15 : 64	A1
		[4]
		Total 10 marks

Useful sketch



Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 8 of 15 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct vector for $\overrightarrow{AB}$
(b)	**Note carefully.** For the first two marks in part (b) you must follow through their working using their $\overrightarrow{AB}$ or $\overrightarrow{AC}$ or $\overrightarrow{BD}$	
	M1	For one correct vector statement for $\overrightarrow{BX}$
	M1	For a second correct and different vector statement for $\overrightarrow{BX}$ For example: $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ OR $\overrightarrow{BX} = \overrightarrow{BC} + \lambda\overrightarrow{AC} = 3\mathbf{a} - \lambda(5\mathbf{a} - 2\mathbf{b})$
	ddM1	For equating coefficients of **a** and **b** and forming two equations in k and another parameter. This must be correct **Dependent on first two M marks**
	dddM1	For solving their simultaneous equations to find k Allow one arithmetical error in processing. **Dependent on all three previous M marks**
	A1	For $k = \frac{3}{8}$
(c)	M1	For using their k to find the ratio of the areas of for example, $CXD : CAD \Rightarrow \frac{CXD}{CAD} = \frac{3}{8}$ or $\frac{CXD}{CAD} = \frac{\frac{1}{2}\times\frac{3}{8}\times 2b\times\sin ACD}{\frac{1}{2}\times\frac{8}{8}\times 2b\times\sin ACD} = \frac{3}{8}$ **OR** $CXD : BCD \Rightarrow \frac{CXD}{BCD} = \frac{5}{8}$ or $\frac{CXD}{CAD} = \frac{\frac{1}{2}\times\frac{5}{8}\times 2b\times\sin BDC}{\frac{1}{2}\times\frac{8}{8}\times 2b\times\sin BDC} = \frac{5}{8}$
	M1	For using the given lengths of BC and AD [3 and 5],to find the ratio of the area of triangle ACD : trapezium $ABCD$ $\frac{CAD}{ABCD} = \frac{\frac{1}{2}\times h\times 5}{\frac{1}{2}\times h\times(5+3)} = \frac{5}{8}$ **OR** $\frac{BCD}{ABCD} = \frac{\frac{1}{2}\times h\times 3}{\frac{1}{2}\times h\times(5+3)} = \frac{3}{8}$
	Note carefully. You must ft their ratios above and check to see if they are being combined correctly for the final M mark here.	
	M1	Combines the areas above to obtain: $\frac{CXD}{CAD}\times\frac{CAD}{ABCD} = \frac{3}{8}\times\frac{5}{8} = \left(\frac{15}{64}\right)$ **OR** $\frac{CXD}{BCD}\times\frac{BCD}{ABCD} = \frac{5}{8}\times\frac{3}{8} = \left(\frac{15}{64}\right)$
	A1	For the correct ratio 15 : 64

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 9 of 15 · find part b in this mark scheme extract

Question	Scheme	Marks
9(a)	$\frac{y-8}{8-3}=\frac{x-6}{6--4}\Rightarrow\frac{y-8}{5}=\frac{x-6}{10}\Rightarrow\left[y-8=\frac{1}{2}(x-6)\right]$	M1A1 [2]
(b)	$C=\left(\frac{4\times6+1\times-4}{4+1},\frac{4\times8+1\times3}{4+1}\right)=(4,7)$	B1,B1 [2]
(c)	$-2=\frac{q-7}{p-4}\Rightarrow q=15-2p$ $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ $\Rightarrow 320=(4-p)^2+(2p-8)^2\Rightarrow p^2-8p-48=0$ $\Rightarrow(p+4)(p-12)=0$ $\Rightarrow p=-4\quad q=15-2\times-4=23\Rightarrow D=(-4,23)$	M1 M1 M1 M1 A1A1 [6]
(d)	Area of triangle $ACD=\frac{1}{2}\times(23-3)\times(4--4)=80$ [square units] **ALT 1** $\frac{1}{2}\begin{bmatrix}-4&4&-4&-4\\3&7&23&3\end{bmatrix}=\frac{1}{2}\left[(-28+92-12)-(12-28-92)\right]=80$ **ALT 2** $A=\frac{1}{2}\times AC\times CD=\frac{1}{2}\times4\sqrt{5}\times8\sqrt{5}=80$	M1A1 [2] [M1A1] [M1A1]
		Total 12 marks

Part	Mark	Notes
(a)	M1	For a **complete** method to find an equation of the line segment AB They must either use the formula shown Award M1A1 for $\frac{y-8}{5}=\frac{x-6}{10}$ (the denominators must be processed). OR They must find the gradient using a correct method and then use the formula $y-y_1=m(x-x_1)\Rightarrow y-8='\frac{1}{2}'(x-6)$ or using $(-4,3)$ OR They must find the gradient using a correct method and then using $y=mx+c$ they must find c and **put an equation together** to score this mark. $y='\frac{1}{2}'x+'5'$ Allow for example $k=\frac{1}{2}x+5$ for this mark (M1) only.
	A1	For a correct equation in any form.[See above]

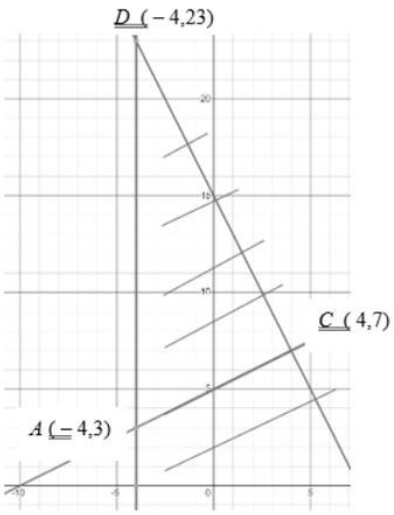
Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 10 of 15 · find part b in this mark scheme extract

(b)	B1	For either correct coordinate of C (4, 7) **This is an M mark in Epen**
	B1	For both correct coordinates of C (4, 7) **This is an M mark in Epen**
(c)	**Note:** You must ft their coords of C in part (c)	
	M1	For using the perpendicular gradient of their equation in (a) to set up a correct equation in terms of p and q **OR** For finding the equation of the line CD using their point C (4, 7) The method must be complete for this mark. $\frac{y-7}{7-23}=\frac{x-4}{4--4} \Rightarrow y=-2x+15$
	M1	For using the given length of CD to set up a second equation using Pythagoras theorem in terms of p and q $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ oe
	M1	For combining their two equations in p and q to obtain a 3TQ
	M1	For solving their 3TQ (provided it is a 3TQ) by any acceptable method. If the 3TQ is incorrect, there must be evidence of a method to score this mark. If they use their calculator on a correct 3TQ and obtain the correct values, score M1.
	A1	For the correct value of p or q
	A1	For the correct values of both p and q
(d)	M1	For any correct method to find the area of triangle ACD, using their coordinates. If they use the determinant method, check their work carefully.
	A1	For the correct area of 80 [square units].

Useful sketch



Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 11 of 15 · find part b in this mark scheme extract

Question	Scheme	Marks
10(a)	$\alpha+\beta=-\frac{k}{2}\qquad \alpha\beta=2$	B1
	$(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta,\quad \alpha^2-\beta^2=(\alpha+\beta)(\alpha-\beta)$	B1B1
	$\Rightarrow \alpha^2-\beta^2=(\alpha+\beta)\sqrt{(\alpha+\beta)^2-4\alpha\beta}$	
	$\Rightarrow \alpha^2-\beta^2=\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times 2}=\frac{7\sqrt{17}}{4}$	M1
	$\Rightarrow \frac{k^4}{4}-8k^2-\frac{833}{4}=0\Rightarrow k^4-32k^2-833=0$ or any equivalent	M1A1
	$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7$ *	M1A1 cso [8]
(b)	Product: $\left[(\alpha-\beta)(\alpha+\beta)=\alpha^2-\beta^2=\frac{7\sqrt{17}}{4}\right]$ from (a)	B1
	Sum: $(\alpha-\beta)+(\alpha+\beta)=\frac{\sqrt{17}}{2}+\frac{7}{2}=\left(\frac{\sqrt{17}+7}{2}\right)$	M1
	$x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0\Rightarrow 4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$	M1A1 [4]
		Total 12 marks

There is an alternative solution using the roots of the equation.

Roots are $\frac{-k\pm\sqrt{k^2-4\times 2\times 4}}{2\times 2}$

$$\Rightarrow\left(\frac{-k+\sqrt{k^2-32}}{4}\right)^2-\left(\frac{-k-\sqrt{k^2-32}}{4}\right)^2=\frac{7\sqrt{17}}{4}$$

$$\Rightarrow\left(-k+\sqrt{k^2-32}\right)^2-\left(-k-\sqrt{k^2-32}\right)^2=28\sqrt{17}$$

$$\Rightarrow k^2-2k\sqrt{k^2-32}+k^2-32-k^2-2k\sqrt{k^2-32}-k^2+32=28\sqrt{17}$$

$$\Rightarrow -4k\sqrt{k^2-32}=28\sqrt{17}\Rightarrow k\sqrt{k^2-32}=7\sqrt{17}$$

$$\Rightarrow k^2\left(k^2-32\right)=833\Rightarrow k^4-32k^2-833=0$$

$$\Rightarrow (k^2-49)(k^2+17)=0\Rightarrow k=-7\ *$$

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 12 of 15 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct sum in terms of k and the correct product. Look out for these embedded in their work.
	B1	For the correct algebra on $(\alpha-\beta)^2=(\alpha+\beta)^2-4\alpha\beta$ Substitution not required for this mark, although it can be implied from correct substitution without the actual algebra seen.
	B1	For the correct algebra on $\alpha^2-\beta^2$ Award also for $(\alpha+\beta)(\alpha-\beta)=\frac{7\sqrt{17}}{4}$ Substitution not required for this mark, although it can be implied from correct substitution without the actual algebra seen.
	M1	For substituting in the values of their sum and product into their $\alpha^2-\beta^2$ This can be implied from sight of $\left(-\frac{k}{2}\right)\sqrt{\left(-\frac{k}{2}\right)^2-4\times2}=\frac{7\sqrt{17}}{4}$
	M1	For forming a 3TQ in terms of k^2 in any form
	A1	For the correct 3TQ Accept it in any form as long as it is 3 terms. For example, accept $\frac{k^4}{16}-2k^2-\frac{833}{16}=0$ etc
	M1	For any acceptable attempt seen to solve their 3TQ [see General Guidance] If there is no working, just $k=-7$ following a 3TQ then award M0 However, accept evidence of $k^2=49$ [with $k^2=-17$]
	A1 cso	For the correct value of k * **Note: This is a given value**
(b)	B1	For a value for the product using their results from (a) or just writes it down.
	M1	For a correct method to find the value of the sum using their values for $(\alpha-\beta)$ and $(\alpha+\beta)$
	M1	For forming a 3TQ with their sum and product. = 0 is not required for the award of this mark
	A1	For a correct equation simplified or unsimplified. For example, accept, $x^2-\left(\frac{\sqrt{17}+7}{2}\right)x+\frac{7\sqrt{17}}{4}=0$ as well as $4x^2-2\left(\sqrt{17}+7\right)x+7\sqrt{17}=0$ o.e.

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 13 of 15 · find part b in this mark scheme extract

Question	Scheme	Marks
11(a)	$f(\theta)=(2\cos\theta-\sin\theta)(2\sin\theta+\cos\theta)$ $=4\sin\theta\cos\theta+2\cos^2\theta-2\sin^2\theta-\sin\theta\cos\theta$ $=3\sin\theta\cos\theta+2(\cos^2\theta-\sin^2\theta)\Rightarrow \frac{3}{2}\sin 2\theta, +2\cos 2\theta$ $=\frac{3}{2}\sin 2\theta+2\cos 2\theta$ *	M1 M1A1 cso [3]
(b)	$\frac{3}{2}\sin 2\theta+2\cos 2\theta+2=0\Rightarrow 3\sin 2\theta+4\cos 2\theta+4=0$ $\Rightarrow 3\sin 2\theta+4(\cos^2\theta-\sin^2\theta)+4(\sin^2\theta+\cos^2\theta)=0$ $\Rightarrow 6\sin\theta\cos\theta+8\cos^2\theta=0$ $\Rightarrow \cos\theta(6\sin\theta+8\cos\theta)=0$ $\Rightarrow \cos\theta=0\Rightarrow\theta=\frac{\pi}{2}$ * $\left[\Rightarrow \tan\theta=-\frac{4}{3}\right]$	M1M1 M1 M1 A1 cso [5]
(c)	$\text{Area}=\int_0^{\frac{\pi}{2}}\left(\frac{3}{2}\sin 2\theta+2\cos 2\theta+2\right)d\theta=\left[-\frac{3}{4}\cos 2\theta+\frac{2\sin 2\theta}{2}+2\theta\right]_0^{\frac{\pi}{2}}$ $=\left(-\frac{3}{4}\cos 2\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+2\left(\frac{\pi}{2}\right)\right)-\left(-\frac{3}{4}\cos 0+\frac{2\sin 0}{2}+2\times 0\right)$ $=\left(\frac{3}{4}+\pi\right)-\left(-\frac{3}{4}\right)=\frac{3}{2}+\pi$	M1 M1 A1 [3]
		Total 11 marks

Part	Mark	Notes
(a)	M1	For multiplying out the brackets **correctly** and simplifying to $k\sin\theta\cos\theta$, $l(\cos^2\theta-\sin^2\theta)$ where k and l are integers. Condone invisible brackets if the working is clear.
	M1	For using the identity for $\cos 2\theta$ on their $2(\cos^2\theta-\sin^2\theta)$ **OR** For using the identity for $\sin 2\theta$ on their $3\sin\theta\cos\theta$
	A1 cso	For the correct identity as shown with no errors. You must check every line of working carefully – this is a given answer.
	Solutions based on RHS = LHS are acceptable. Apply the above marks – if you are not sure, please send to REVIEW	

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 14 of 15 · find part b in this mark scheme extract

(b)	M1	For correctly using the identity $\cos 2\theta = \cos^2\theta - \sin^2\theta$
	M1	For correctly using the identity $\sin^2\theta + \cos^2\theta = 1$
	M1	For correctly using the identity $\sin 2\theta = 2\sin\theta\cos\theta$
	M1	For factorising the resulting expression. You must see this step
	A1 cso	For $\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$
	ALT	
	M1	For correctly using $\cos 2\theta = \cos^2\theta - \sin^2\theta$ $\frac{3}{2}\sin 2\theta + 2\cos 2\theta + 2 = \frac{3}{2}\sin 2\theta + 2\left(\cos^2\theta - \sin^2\theta\right) = 0$
	M1	For correctly using the identity $\sin^2\theta + \cos^2\theta = 1$ $\frac{3}{2}\sin 2\theta + 2\cos 2\theta + 2 = \frac{3}{2}\sin 2\theta + 2\left(\cos^2\theta - 1 + \cos^2\theta\right) + 2 = 0$ $\Rightarrow \left[\frac{3}{2}\sin 2\theta + 4\cos^2\theta = 0\right]$
	M1	For correctly using the identity $\sin 2\theta = 2\sin\theta\cos\theta$ $\frac{3}{2}\sin 2\theta + 4\cos^2\theta = 3\sin\theta\cos\theta + 4\cos^2\theta = 0$
	M1	For factorising the resulting expression. You must see this step $3\sin\theta\cos\theta + 4\cos^2\theta = \cos\theta\left(3\sin\theta + 4\cos\theta\right) = 0$
	A1	$\cos\theta = 0 \Rightarrow \theta = \frac{\pi}{2}$ *
(c)	M1	For integrating the given expression. Minimally acceptable integration as shown below: $\frac{3}{2}\sin 2\theta \Rightarrow \pm\frac{3}{4}\cos 2\theta$ $2\theta\cos 2\theta \Rightarrow \pm\left(\frac{2}{2}\right)\sin 2\theta$ $2 \Rightarrow 2\theta$
	M1	For substituting **BOTH** of the given limits the correct way around into **their changed expression** and subtracting.. There must be a minimum of two terms to substitute limits into. This must be explicitly seen
	A1	For the correct exact area as shown. [Approximate area = 4.641….]

Type MS · Quadratic inequalities

Answer 5 of 10 · Question 4 part b · 4 marks

2024 JUN · Paper 01 (regional R) · extract 15 of 15 · find part b in this mark scheme extract

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Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 1 of 23 · find part b in this mark scheme extract

2(a)	$x < \frac{1}{2}$ o.e	B1 (1)
(b)	$10x^2 + 7x - 12 (< 0)$ $(5x - 4)(2x + 3) (< 0)$ $x = \frac{4}{5}$ $\quad$ $x = -\frac{3}{2}$	M1
	$-\frac{3}{2} < x < \frac{4}{5}$ o.e	M1A1 (3)
(c)	$-\frac{3}{2} < x < \frac{1}{2}$	B1FT (1)
		Total 5 marks

Part	Mark	Additional Guidance
(a)	B1	$x < \frac{1}{2}$ o.e
(b)	M1	Rearranges to a 3TQ and attempts to solve. See general guidance for what constitutes an attempt to solve. If they use a calculator to solve the 3TQ, the 3TQ must be correct and the values of x also correct. They must obtain **two** values of x for this mark
	M1	For forming an **inside** region with their critical values.
	A1	For the correct inequality. May be seen described in set notation.
(c)	B1FT	For $-\frac{3}{2} < x < \frac{1}{2}$ FT their (a) and (b) provided they overlap and it is an inside region. **NB** The final answer given must be consistent with their parts (a) and (b)

Question	Scheme	Marks

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 2 of 23 · find part b in this mark scheme extract

number		

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 3 of 23 · find part b in this mark scheme extract

M1	Multiplies numerator and denominator by $6+2\sqrt{5}$ with no errors
M1	Equates numerator coefficients to find two simultaneous equations in a and b
M1	Solves their simultaneous equations by any method provided the equations of the form; $ma+nb=...$ and $pa+qb=...$ where $m \neq p$ and $n \neq q$ They must reach both a value for a and a value for b for this mark
M1	For multiplying through by 8 or 4 to give a prime number in the denominator of $\frac{9+4\sqrt{5}}{c}$
A1	For the correct values of a, b and c Allow embedded values.
ALT	
M1	'Cross multiplies' through by $\left(6-2\sqrt{5}\right)$ **AND** by c
M1	Multiplies out both sides with no errors
M1	Collects up like terms to obtain $c\left(a+b\sqrt{5}\right)=A+B\sqrt{5}$
M1	Factorises their expression to obtain $P\left(Q+R\sqrt{5}\right)$ where P, Q and R are integers or deduces by inspection that $a = 7$ and $c = 2$
A1	For the correct values – accept if embedded in the expression.

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 4 of 23 · find part b in this mark scheme extract

Question number	Scheme	Marks
4 (a) (i)	$\frac{ar^6}{ar^2}=\frac{40}{10}$	M1
	$r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$	M1
	$r=-\sqrt{2}$	A1
(ii)	$ar^2=10\Rightarrow a=\frac{10}{'2'}$	M1
	$a=5$	A1 (5)
(b)	$\lvert r\rvert>1$ so the series is not convergent.	B1FT (1)
		Total 6 marks

Part	Mark	Additional Guidance
(a)(i)	M1	For $\frac{ar^6}{ar^2}=\frac{40}{10}$ or $\frac{ar^2}{ar^6}=\frac{40}{10}$
	M1	For rearranging to find a value for r $r=\sqrt[4]{\frac{40}{10}}=(\pm\sqrt{2})$ or $\left(\pm\sqrt[4]{4}\right)$
	A1	For the correct value of $r=-\sqrt{2}$ Accept $-\sqrt[4]{4}$
(a)(ii)	M1	For attempting to find a value for a using their r $ar^2=10\Rightarrow a=\frac{10}{'2'}$
	A1	For the correct value of $a=5$ Allow this mark from the use of $r=\sqrt{2}$
(b)	B1FT	$\lvert r\rvert>1$ so the series is not convergent. Follow through their value of r provided it is greater than 1.

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 5 of 23 · find part b in this mark scheme extract

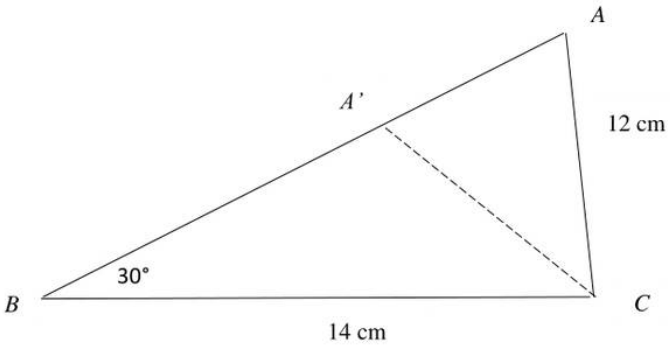
Question number	Scheme	Marks
5(a)	$12^2 = 14^2 + x^2 - 2\times 14x\cos 30$	M1
	$x^2 - 14\sqrt{3}x + 52 = 0$	M1 A1
	$x = \dfrac{14\sqrt{3} \pm \sqrt{(14\sqrt{3})^2 - 4(1)(52)}}{2(1)}$	M1
	$x = 7\sqrt{3} \pm \sqrt{95}$	A1cso [5]
(b)	Area $= \dfrac{1}{2}\times 14\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \sin 30$	M1
	or uses Area $= \dfrac{1}{2}\times 12\times\left(7\sqrt{3} \pm \sqrt{95}\right)\times \dfrac{7}{12}$	
	Difference in areas	
	$D = \dfrac{1}{2}\times 14\times\left(7\sqrt{3} + \sqrt{95}\right)\times \sin 30 - \dfrac{1}{2}\times 14\times\left(7\sqrt{3} - \sqrt{95}\right)\times \sin 30$	dM1
	$D = 7\sqrt{95}$	A1 [3]
ALT Part (a) – sine rule		
	$\dfrac{12}{\sin 30} = \dfrac{14}{\sin A} \Rightarrow \sin A = \dfrac{7}{12}$	{M1
	$\left(\dfrac{7}{12}\right)^2 + \cos^2 A \equiv 1 \Rightarrow \cos A = \sqrt{1 - \left(\dfrac{7}{12}\right)^2}$	M1
	$\sin A = \dfrac{7}{12}, \cos A = \pm\dfrac{\sqrt{95}}{12}$	A1
	$x = 12\cos A + 14\cos 30 \Rightarrow x = 12\times\dfrac{\pm\sqrt{95}}{12} + 14\times\dfrac{\sqrt{3}}{2}$	M1
	$x = 7\sqrt{3} \pm \sqrt{95}$ *	A1 Cso [5]}
		Total 8 marks

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 6 of 23 · find part b in this mark scheme extract

USEFUL SKETCH



Part	Mark	Additional Guidance
(a)	M1	Use the cosine rule to form an equation. Rule to be correct but may be in any rearranged form.
	M1	Rearrange to obtain a 3TQ. Terms in any order. = 0 may be missing.
	A1	Correct 3TQ Allow poor notation – missing = 0
	M1	Attempt to solve their 3TQ by use of the formula or completing the square. See general guidance for what constitutes an attempt to solve. Use of calculators: If the 3TQ is correct, accept correct solutions without working.
	A1	Correct values for x $x = 7\sqrt{3} \pm \sqrt{95}$
(b)	M1	For use of area of triangle $= \frac{1}{2}ac\sin B$ with $a = 14$, $B = 30°$ and *their* x May use either value found for x. Or uses $= \frac{1}{2}bc\sin A$ with $c = 12$ and $\sin A = \frac{7}{12}$
	dM1	For finding the difference between their two areas provided both areas are positive This M mark is dependent on the previous M mark
	A1	Correct difference in areas (must be exact value)

	ALT Part (a) – Uses sine rule	
	M1	Use of the sine rule to find sin$A$
	M1	Use of $\sin^2 A + \cos^2 A \equiv 1$ to find an exact value for $\cos A$
	A1	Correct exact values for sin$A$ and $\cos A$
	M1	Method to find x $x = 12\cos A + 14\cos 30$ Allow *their* $\cos A$
	A1	Correct values for x $x = 7\sqrt{3} \pm \sqrt{95}$

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 7 of 23 · find part b in this mark scheme extract

Question number	Scheme	Marks
6 (a)	$700\pi = 2\pi r^2 + 2\pi rh$	B1
	$h = \dfrac{350 - r^2}{r}$ or $rh = 350 - r^2$	M1
	$V = \pi r^2 h = \pi r^2\left(\dfrac{350 - r^2}{r}\right) = \left\{\pi r\left(350 - r^2\right)\right\} = \pi r\left(350 - r^2\right)$*	M1A1* cso [4]
(b)	$V = 350\pi r - \pi r^3$	
	$\dfrac{dV}{dr} = 350\pi - 3\pi r^2 = 0$	M1
	$\Rightarrow 350\pi = 3\pi r^2 \Rightarrow r = \sqrt{\dfrac{350}{3}} = 10.8$ (3sf)	M1A1
	$\dfrac{d^2V}{dr^2} = -6\pi r$	M1
	$\dfrac{d^2V}{dr^2} < 0 \therefore$ max	A1 [5]
(c)	$h = \dfrac{350 - r^2}{r} = \dfrac{350 - \dfrac{350}{3}}{\sqrt{\dfrac{350}{3}}} = 21.6$ (3 s.f.)	B1 [1]
		Total 10 marks

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 8 of 23 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For a **correct** equation for the surface area in terms of the radius and height
	M1	For rearranging to find h in terms of r which must be as a minimum $h = \frac{k - lr^2}{mr}$ where k, l and m are integers OR $rh = k - lr^2$ where k and l are integers
	M1	For a correct expression for volume using *their* h in terms of r
	A1* cso	For rearranging and obtaining the given result with no errors or omissions and must include $V = \ldots$ [Allow Volume = …. or Vol = …..]
(b)	M1	For an attempt to differentiate the **given** expression for V which must be of the form $\frac{dV}{dr} = 350\pi - K\pi r^2$ where K is a constant Using Product Rule: $\frac{dV}{dr} = \pi r(-2r) + \pi(350 - r^2) \Rightarrow \frac{dV}{dr} = \left[350\pi - K\pi r^2\right]$
	M1	For setting their $\frac{dV}{dr} = 0$ **and** attempting to find a value for r.
	A1	$r = \sqrt{\frac{350}{3}}$ [=10.801...] accept awrt 10.8
	M1	For finding $\frac{d^2V}{dr^2}$ Accept only $\frac{d^2V}{dr^2} = \pm k\pi r$ where k is a constant See general guidance for what constitutes an attempt to differentiate.
	A1	For the correct conclusion derived from the correct working throughout.
(c)	B1	For obtaining the value of h correct to 3 s.f. Accept awrt 21.6

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 9 of 23 · find part b in this mark scheme extract

<table>
<tr><th>Question number</th><th>Scheme</th><th>Marks</th></tr>
<tr><td>7 (a)</td><td>
<table>
<tr><td>x</td><td>0</td><td>0.5</td><td>1</td><td>1.5</td><td>2</td><td>2.5</td><td>3</td><td>3.5</td></tr>
<tr><td>y</td><td>4.26</td><td>4.14</td><td>4</td><td>3.83</td><td>3.63</td><td>3.37</td><td>3</td><td>2.37</td></tr>
</table>
</td><td>B2
(2)</td></tr>
<tr><td>(b)</td><td>Points plotted within half a square.
Points joined with a smooth curve.</td><td>B1ft
B1ft
(2)</td></tr>
<tr><td>(c)</td><td>$3^{2x-5}=(4-x)^3 \Rightarrow 2x-5=3\log_3(4-x)$
$\frac{2x-5}{3}+3=\log_3(4-x)+3 \Rightarrow \frac{2x+4}{3}=\log_3(4-x)+3$
ALT
$\log_3(4-x)+3=mx+c$
$\Rightarrow \log_3(4-x)=mx+(c-3)$
$\Rightarrow (4-x)=3^{mx+(c-3)}$
$\Rightarrow (4-x)^3=3^{(mx+(c-3))3}=3^{3mx+3c-9}$
$\Rightarrow 2x-5=3mx+3c-9 \Rightarrow 3m=2$ and $-5=3c-9$
$\Rightarrow m=\frac{2}{3} \quad c=\frac{4}{3}$
Graph of $y=\frac{2x+4}{3}$ drawn. Intersection point is $x=2.8$</td><td>M1M1
M1A1
[M1
M1
M1
A1]
M1A1
(6)
[10]</td></tr>
</table>

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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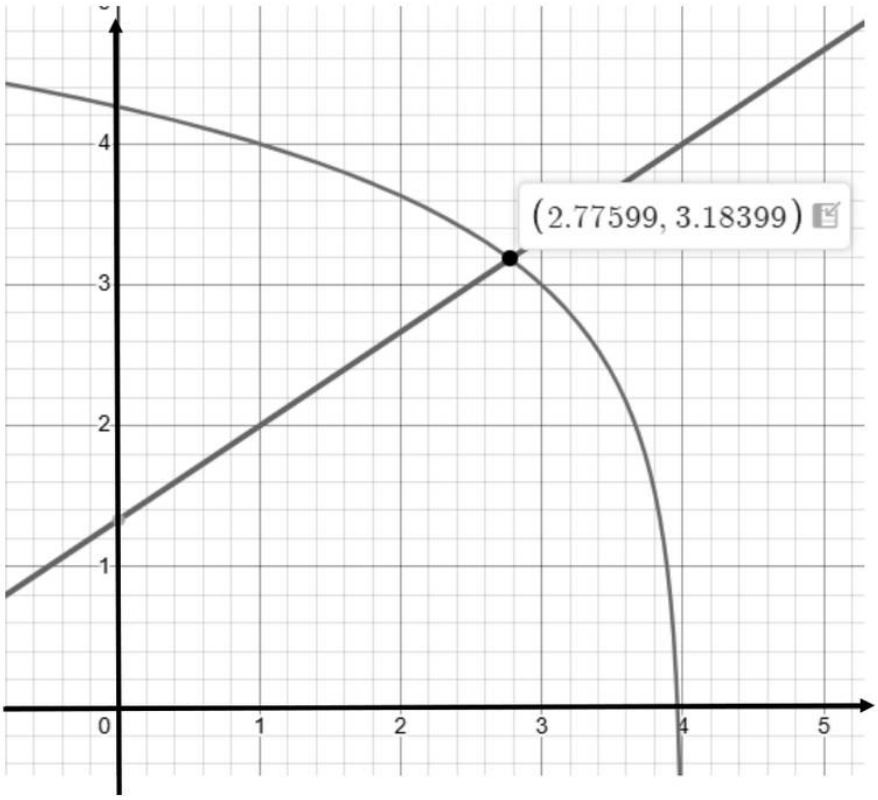
Part	Mark	Additional Guidance
(a)	B2	All three points correct (B1 for 2 points correct)
(b)	B1ft	Points plotted ft their table. Allow half a square tolerance.
	B1ft	Points joined together with a smooth curve. Allow half a square tolerance on joining up their points. FT their table
(c)	M1	For taking logs base 3 of both sides $2x-5=\log_3(4-x)^3$
	M1	For applying the power law $2x-5=3\log_3(4-x)$
	ddM1	For attempting to rearrange to the required form An attempt is defined as dividing through by 3 and adding 3 to both sides.
	A1	For the correct line, in any form $\frac{2x-5}{3}+3=\log_3(4-x)+3\Rightarrow\left[\left(y=\right)\frac{2}{3}x+\frac{4}{3}\right]$
	ALT for first 4 marks	
	M1	Sets up the equation to obtain $\log_3(4-x)+3=mx+c$ And removes the log to obtain $(4-x)=3^{mx+(c-3)}$
	M1	Raises **both** sides to a power of 3. Allow one error in this on the LHS
	M1	Compares coefficients to obtain an equation in m and an equation in c
	A1	Obtains the values $m=\frac{2}{3}$ $c=\frac{4}{3}$
	Draws the line	
	M1	For their line drawn provided it is of the form $y=kx\pm\frac{4}{3}$ for $k\neq 0,\ 1$
	A1	$x=2.8$ or 2.7 Calculator value is 2.776

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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USEFUL GRAPH



Type MS · Quadratic inequalities

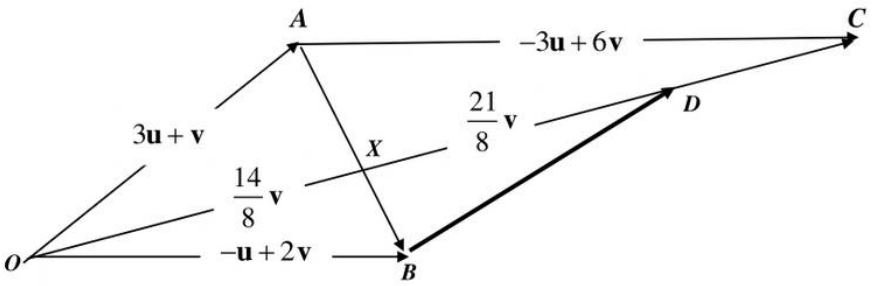
Answer 6 of 10 · Question 2 part b · 3 marks

2025 JUN · Paper 02 · extract 12 of 23 · find part b in this mark scheme extract

Question number	Scheme	Marks
8 (a)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}, = -(3\mathbf{u}+\mathbf{v}) + (2\mathbf{v}-\mathbf{u}) = -4\mathbf{u}+\mathbf{v}$	M1,A1 (2)
(b)	$\overrightarrow{AC} = \lambda(2\mathbf{v}-\mathbf{u})$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = (3\mathbf{u}+\mathbf{v}) + \lambda(2\mathbf{v}-\mathbf{u}) = \mu\mathbf{v}$	M1
	$\mathbf{u}: 0 = 3-\lambda \qquad \mathbf{v}: \mu = 1+2\lambda$	M1
	$\mu = 7 \qquad [\lambda = 3]$	A1 (4)
(c)	$\overrightarrow{OX} = \overrightarrow{OA} + \alpha\overrightarrow{AB} = 3\mathbf{u}+\mathbf{v}+\alpha(-4\mathbf{u}+\mathbf{v})$ or $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AC} + \beta\overrightarrow{CO} = 3\mathbf{u}+\mathbf{v}+3(2\mathbf{v}-\mathbf{u})+\beta(-7\mathbf{v})$	M1
	$\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$	A1
	$\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$	M1
	$\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$	M1
	$\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{35}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{7}{4}\mathbf{v}+\frac{21}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}+\frac{19}{8}\mathbf{v}$	A1 (5)
		Total 11 marks

USEFUL SKETCH

$$\overrightarrow{OD} = \frac{35}{8}\mathbf{v} \quad \text{and} \quad \overrightarrow{AX} = -3\mathbf{u}+\frac{3}{4}\mathbf{v}$$



Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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Part	Mark	Additional Guidance
(a)	M1	Use of $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$
	A1	$-4\mathbf{u} + \mathbf{v}$
(b)	M1	For use of $\overrightarrow{AC}$ is parallel to $\overrightarrow{OB}$ to form an expression for $\overrightarrow{AC}$
	M1	For use of their expression for $\overrightarrow{AC}$ to form an expression for $\overrightarrow{OC}$
	M1	For equating for **u** and for **v**
	A1	For $\mu = 7$
(c)		**General principles for marking part (c)** For example: You may see the vector path for $\overrightarrow{BD}$ first - score the 4th M mark • Candidates will need to find vector $\overrightarrow{OX}$ [M1A1] • Candidates will need to use the ratio of bases to find $\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$ **leading** to establish $\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$ [M1] • Check any path they choose on the diagram below. For example, $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OX} + \overrightarrow{XD}$ or [M1] $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OA} + \overrightarrow{AX} + \overrightarrow{XD}$ If a candidate chooses to take a longer path and need to find for example, $\overrightarrow{AX}$ they will need to find a vector. There is no separate mark for this. • For the correct vector
	M1	For an expression for $\overrightarrow{OX}$ including an unknown constant
	A1	For $\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$
	M1	For recognising that triangle OBX and triangle OBD share a base OB and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{OD}$ together with an attempt to find $\overrightarrow{OD}$ OR For recognising that triangle OBX and triangle BXD share a base BX and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{XD}$ together with an attempt to find $\overrightarrow{XD}$
	M1	For method to find $\overrightarrow{BD}$ Award this for any valid path given
	A1	$\overrightarrow{BD} = \mathbf{u} + \frac{19}{8}\mathbf{v}$

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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Question number	Scheme	Marks
9 (a)	$\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ *	M1A1A1 Cso [3]
(b)	**Method A**	
	$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$	M1
	$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$	M1
	$2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$	M1
	$\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y \Rightarrow \frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$ $\Rightarrow \frac{d^2y}{dt^2} = -8\frac{dy}{dt} - 20y \Rightarrow \frac{d^2y}{dt^2} + 8\frac{dy}{dt} + 20y = 0$	M1
	$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0 \Rightarrow M = 8, N = 20$	A1 [5]
	Method B	
	$\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$ $\left(= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t\right)$	[M1
	$\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$	
	$= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$	M1
	$= (16 - 2M)e^{-4t}\sin 2t + (12 - 4M + N)e^{-4t}\cos 2t$	M1
	$16 - 2M = 0$ and $12 - 4M + N = 0$	M1
	$M = 8,\ N = 20$	A1]
	Method C By inspection – please see additional guidance.	
		Total 8 marks

Part	Mark	Additional Guidance

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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(a)	M1	For an attempt to differentiate the given expression. • There needs to be an acceptable attempt to differentiate both terms. $e^{-4t}\cos 2t \rightarrow ke^{-4t}\sin 2t + le^{-4t}\cos 2t$ with $k, l \neq 0$ • There need to be two terms added. $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t$
	A1	One term completely correct. $\frac{dy}{dt} = -4e^{-4t}\cos 2t + '-2e^{-4t}\sin 2t$ OR $\frac{dy}{dt} = '-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t$
	A1	Fully correct differentiated expression and rearranged into the required form $\frac{dy}{dt} = -4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t \Rightarrow 2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ Accept also, For example: $-\left(-4e^{-4t}\cos 2t' - 2e^{-4t}\sin 2t\right) - 4\left(e^{-4t}\sin 2t\right) = 2e^{-4t}\sin 2t$ With any unambiguous indication that a conclusion has been reached, e.g., QED, hash sign, double underline, RHS = LHS etc.
		Method A
(b)	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4x}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For substituting y and $\frac{dy}{dt}$ into their $\frac{d^2y}{dt^2}$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + 8e^{-4t}\sin 2t - 4y$
	M1	For preparing to eliminate $\cos 2t$ by rearranging their $\frac{dy}{dt}$ $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4e^{-4t}\cos 2t \Rightarrow 8e^{-4t}\sin 2t = -4\frac{dy}{dt} - 16y$ Allow errors in arithmetic but not mathematically incorrect process.
	M1	For an unsimplified expression only in terms of y, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ $\frac{d^2y}{dt^2} = -4\frac{dy}{dt} + \left\{-4\frac{dy}{dt} - 16y\right\} - 4y$
	A1 [8]	For finding the value of M and the value of N. $M = 8,\ N = 20$
		Method B

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

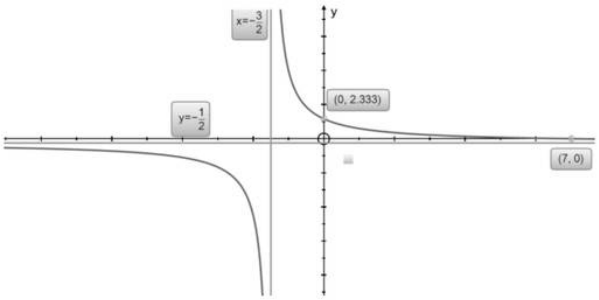
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	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4t}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For substituting y, $\frac{dy}{dt}$ and $\frac{d^2y}{dt^2}$ into the equation. $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny$ $= 16e^{-4t}\sin 2t + 12e^{-4t}\cos 2t + M(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + Ne^{-4t}\cos 2t$
	M1	For collecting terms $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = ('16' - 2M)e^{-4t}\sin 2t + ('12' - 4M + N)e^{-4t}\cos 2t$
	M1	For identifying two equations from their working and a method to solve them simultaneously For example, $'16' - 2M = 0$ and $'12' - 4M + N = 0$
	A1	For finding the value of M and the value of N. $M = 8, N = 20$
		Method C (By inspection)
	M1	For an attempt to find $\frac{d^2y}{dt^2}$ Minimally acceptable attempt is $ke^{-4t}\sin 2t + le^{-4t}\cos 2t \rightarrow k(me^{-4t}\sin 2t + ne^{-4t}\cos 2t) + l(pe^{-4t}\cos 2t + qe^{-4t}\sin 2t)$ k, l as in their first derivative, $m, n, p, q \neq 0$ $\frac{d^2y}{dt^2} = -4(-4e^{-4t}\cos 2t - 2e^{-4t}\sin 2t) + 8e^{-4t}\sin 2t - 4e^{-4t}\cos 2t$
	M1	For simplifying $\frac{d^2y}{dt^2} = e^{-4t}16\sin 2t + e^{-4t}12\cos 2t = \left[e^{-4t}(16\sin 2t + 12\cos 2t)\right]$
	M1	For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}[\sin 2t]$ and deducing that $'16' + (\underline{8} \times -2) = 0 \Rightarrow M = 8$
	M1	For comparing $\frac{d^2y}{dt^2}$ with $\frac{dy}{dt}$ and y and deducing that $'12' - ('8' \times 4) + \underline{20} = 0 \Rightarrow N = 20$
	A1	For both $M = 8$ and $N = 20$
		Total 8 marks

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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Question number	Scheme	Marks
10 (a)	(i) $x=-\frac{3}{2}$ (ii) $y=-\frac{1}{2}$	B1 B1 [2]
(b)	$(7,0)$ $\left(0,\frac{7}{3}\right)$	B1 B1 [2]
(c)		B1ft curve B1ft asymptotes B1ft (intersections with x- and y-axes) [3]
(d)	$\frac{-1(2x+3)-2(7-x)}{(2x+3)^2}=\left[\frac{-17}{(2x+3)^2}\right]$	M1
	$\frac{-1\left(2\times\frac{1}{2}+3\right)-2\left(7-\times\frac{1}{2}\right)}{\left(2\times\frac{1}{2}+3\right)^2}=-\frac{17}{16}$	M1A1* cso [3]
	ALT – product rule	
	$-1(2x+3)^{-1}+(7-x)(-1)(2)(2x+3)^{-2}=\left[\frac{-17}{(2x+3)^2}\right]$	M1
	$-1\left(2\times\frac{1}{2}+3\right)^{-1}+\left(7-\frac{1}{2}\right)(-1)(2)\left(2\times\frac{1}{2}+3\right)^{-2}=-\frac{17}{16}$	M1A1* cso

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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(e)	$-\frac{17}{(2x+3)^2}=-\frac{17}{16}$	M1
	$(2x+3)^2=16\Rightarrow 2x+3=\pm\sqrt{16}\Rightarrow x=0.5 \text{ or } -3.5$	dM1
	$y=\frac{7--3.5}{2\times-3.5+3}\left(=-\frac{21}{8}\right)$	ddM1
	$\left(-\frac{7}{2},-\frac{21}{8}\right)$	A1
	$y--\frac{21}{8}=-\frac{17}{16}\left(x--\frac{7}{2}\right)$	M1A1
	$34x+32y+203=0$	A1 [7]
		Total 17 marks

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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Part	Mark	Additional Guidance
For parts (a) and (b) look for their responses in the body of the question. In part (a), if the responses are not labelled (i) and (ii) accept them in order.		
(a)	(i) B1	For $x=-\frac{3}{2}$ oe
	(ii) B1	For $y=-\frac{1}{2}$ oe
(b)	B1 B1	First B1 for either correct, second B1 for both correct Condone if not given as coordinates e.g. $x=7$ and/or $y=\frac{7}{3}$ given
(c)	B1ft	Two branches drawn in the correct two "quadrants" created by the two asymptotes. Mark intention, allow poor curves, but do not allow the curve to bend back on itself or touch any asymptotes.
	B1ft	Two clearly marked asymptotes, ft their (a), labelled as described, there must be one section of the curve present, tending towards these asymptotes.
	B1ft	Two clearly labelled intersections with the axes, ft their (b), at least one section of their curve must pass through one of these intersections. Intersections must be labelled correct way around. If additional intersections seen then B0
(d)	M1	Attempt the quotient rule. Numerator must be of the form $A.(2x+3)-B.(7-x)$, $B>1$. Denominator must be of the form $(2x+3)^2$
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
	ALT – product rule	
	M1	For an attempt at Product Rule. Must be a sum of two products. Must have the form $c(2x+3)^{-1}+d(7-x)(2x+3)^{-2}$ for constants c, d.
	M1	Substitutes $x=\frac{1}{2}$ into their derivative **NB: This is an A mark in Epen**
	A1* cso	Obtains the required value of gradient $\left(-\frac{17}{16}\right)$ with no errors **NB: This is a B mark in Epen**
(e)	M1	For equating the gradient of the tangent to their $\frac{dy}{dx}$ from (d)
	dM1	For rearranging to obtain $(2x+3)^2=16$ and attempt to solve or for rearrangement to a 3TQ and method to solve the 3TQ. See general guidance for acceptable methods to solve.

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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	ddM1	For substitution of their x into the original equation to obtain y. This must **not** be $x=\frac{1}{2}$ This M mark is dependent on the **two** previous M marks
	A1	For correct coordinate of $P\left(-\frac{7}{2},-\frac{21}{8}\right)$
	M1	For substitution of their coordinates of P and the gradient $-\frac{17}{16}$ to obtain an equation of a straight line using any valid method. If $y=mx+c$ is used a value of c must be found **and** an equation needs to be formed. For example; $'-\frac{21}{8}'=-\frac{17}{16}\times'-\frac{7}{2}'+c\Rightarrow c=-\frac{203}{32}\Rightarrow y=-\frac{17}{16}x-\frac{203}{32}$ [
	A1	For any correct equation of the tangent with the correct values of x and y in any form.
	A1	For the correct equation of tangent given in the required form. $34x+32y+203=0$

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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Question number	Scheme	Marks
11 (a)	$\cos^4\theta-\sin^4\theta=(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$	M1
	$=(\cos^2\theta-\sin^2\theta)\times1=\cos2\theta$ *	M1M1A1 cso [4
	ALT example	
	$\cos^4\theta-\sin^4\theta=\cos^4\theta-\sin^2\theta\sin^2\theta$	
	$=\cos^4\theta-(1-\cos^2\theta)(1-\cos^2\theta)$	2nd M1
	$=\cos^4\theta-(1-2\cos^2\theta+\cos^4\theta)$	1st M1
	$=(1-2\cos^2\theta)$	3rd M1
	$=\cos2\theta$	A1
(b)	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-3=2[\cos^4\left(\theta+\frac{\pi}{8}\right)-\sin^4\left(\theta+\frac{\pi}{8}\right)]$	M1
	$=2\cos\left(2\theta+\frac{\pi}{4}\right)$	A1
	$8\cos^2\left(2\theta+\frac{\pi}{4}\right)-2\cos\left(2\theta+\frac{\pi}{4}\right)-3=0$	M1
	$\left(4\cos\left(2\theta+\frac{\pi}{4}\right)-3\right)\left(2\cos\left(2\theta+\frac{\pi}{4}\right)+1\right)[=0]$	
	$\cos\left(2\theta+\frac{\pi}{4}\right)=\frac{3}{4}\quad\cos\left(2\theta+\frac{\pi}{4}\right)=-\frac{1}{2}$	A1
	$2\theta+\frac{\pi}{4}=0.72273...\quad2\theta+\frac{\pi}{4}=\frac{2\pi}{3}$	dM1
	$\theta=2.39,3.11,\ 0.65,1.70$ accept $\theta=\frac{5\pi}{24},\frac{13\pi}{24}$	A1A1 [7]
(c)	$\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos^4 2x-\sin^4 2x-8\sin 4x\right)\mathrm{d}x=\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\left(\cos 4x-8\sin 4x\right)\mathrm{d}x$	M1
	$=\left[\frac{1}{4}\sin 4x+\frac{8}{4}\cos 4x\right]_{\frac{\pi}{16}}^{\frac{\pi}{8}}$	M1
	$=\frac{1}{4}\left[\sin\left(4\times\frac{\pi}{8}\right)-\sin\left(4\times\frac{\pi}{16}\right)\right]+2\left[\cos\left(4\times\frac{\pi}{8}\right)-\cos\left(4\times\frac{\pi}{16}\right)\right]$	M1
	$=\frac{1}{4}\left[1-\frac{\sqrt{2}}{2}\right]+2\left[0-\frac{\sqrt{2}}{2}\right]=\frac{1}{4}-\frac{9\sqrt{2}}{8}$	A1 [4]
		Total 15 marks

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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Part	Mark	Additional Guidance
General Principles for marking part (a) • First M mark is for either applying the difference of two squares or multiplying out their expression and collecting terms. • Second M mark is for applying $\cos^2\theta+\sin^2\theta=1$ in any form. This can be implied from correct work. However, do not award for just $\cos^4\theta-\sin^4\theta=\cos^2\theta-\sin^2\theta$ seen. • Third M mark is for applying any identity for $\cos 2\theta$ • A1 For obtaining the required result with no errors.		
(a)	M1	For factorising as the difference of two squares $(\cos^2\theta-\sin^2\theta)(\cos^2\theta+\sin^2\theta)$
	M1	For use of $\cos^2\theta+\sin^2\theta)=1$
	M1	For use of $\cos^2\theta-\sin^2\theta=\cos(2\theta)$
	A1 cso	Obtain the given result with no errors in the working.
(b)	M1	For using the given result to obtain a quadratic in $\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	A1	For rearranging to the correct quadratic
	M1	For a valid attempt to solve their **3TQ**. See General Guidance for what constitutes an attempt to factorise. If candidates use a calculator [so no working shown] then award this mark only if the 3TQ is correct with the correct solutions.
	A1	For obtaining correct values for $\cos\left(2\theta+\frac{\pi}{4}\right)$ Note: another variable may be used in place of $\left(2\theta+\frac{\pi}{4}\right)$ for this mark
	dM1	For achieving any valid value **in radians** for their $2\theta+\frac{\pi}{4}$ i.e. 0.7227…or $\frac{2}{3}\pi$ Accept values out of range for this mark. This is dependent on the previous M mark.
	A1	For two correct solutions in range
	A1	For all four correct solutions in range and no others. Extra solutions within range scores A0 Ignore solutions outside of range. Exact solutions when given must be given as multiples of π.
(c)	M1	For use of the given result to obtain $\int_{\frac{\pi}{16}}^{\frac{\pi}{8}}\cos 4x\ \mathrm{d}x$
	M1	For an attempt to integrate $\cos kx\rightarrow l\sin kx$ where $k,l\neq 1$
	M1	For substituting limits into a changed expression
	A1	For the correct answer in the required form.

Type MS · Quadratic inequalities

Answer 6 of 10 · Question 2 part b · 3 marks

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Type MS · Quadratic inequalities

Answer 7 of 10 · Question 1 part b · 3 marks

2021 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Number		
1 (a)	$3x < 12$	M1
	$x < 4$	A1
		[2]
(b)	$(2x+1)(x-3) > 0$	M1
	Critical values are $x = -\frac{1}{2}$ and $x = 3$	M1
	$x < -\frac{1}{2}$ $x > 3$	A1
		[3]
(c)	$x < -\frac{1}{2}$ $3 < x < 4$	B1ft
		[1]
		Total 6 marks

Type MS · Quadratic inequalities

Answer 7 of 10 · Question 1 part b · 3 marks

2021 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Guidance
(a)	M1	Attempts to solve the inequality to achieve $3x < 12$ Allow $3x < a$ where a is an integer
	A1	For $x < 4$
(b)	M1	Attempts to solve the inequality by any method to find critical values See General Guidance for acceptable methods. If a calculator is used, the critical values must be fully correct for this mark. Allow = or > for this mark or even no sign at all provided it is clear they are solving a quadratic. $(2x+1)(x-3) > 0 \Rightarrow x = ...,\ ...$ $\left(x = -\frac{1}{2},\ 3\right)$
	M1	For forming a correct inequality, which must be an open interval, following through their **two** critical values which must have come from the solution of a 3TQ. $x < -\frac{1}{2} \quad x > 3$ Accept any correct notation. E.g., $x < -\frac{1}{2}$ or $x > 3$ Or $\left\{x : x < -\frac{1}{2}\right\} \cup \{x : x > 3\}$ Condone $x < -\frac{1}{2}$ and $x > 3$ for this mark only
	A1	For the correct inequality with the correct critical values using any acceptable notation. Eg, $x < -\frac{1}{2} \quad x > 3$
(c)	B1ft	For $x < -\frac{1}{2} \quad 3 < x < 4$ ft their answers from parts (a) and (b) provided (b) is of the form $x < p$ and/or $x > q$ **Note:** If you have already penalised them for writing $x < -\frac{1}{2}$ **and** $x > 3$ in part (b) then allow $x < -\frac{1}{2}$ **and** $3 < x < 4$ for this mark.

Type MS · Quadratic inequalities

Answer 7 of 10 · Question 1 part b · 3 marks

2021 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Quadratic inequalities

Answer 8 of 10 · Question 2 part b · 3 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

2(a)	$2(x+1)<5x-2 \Rightarrow 2x+2<5x-2 \Rightarrow 3x>4 \Rightarrow x>\frac{4}{3}$	M1A1 [2]
(b)	$3x^2-x-10=0 \Rightarrow (3x+5)(x-2)=0 \Rightarrow x=-\frac{5}{3}, 2$ $'-\frac{5}{3}' \leqslant x \leqslant '2'$ allow $'-\frac{5}{3}' < x < '2'$ $-\frac{5}{3} \leqslant x \leqslant 2$	M1 dM1 A1 [3]
(c)	$"\frac{4}{3}" < x \leqslant "2"$	B1ft [1]
		Total 6 marks

Part	Mark	Notes
(a)	M1	Attempts to solve the inequality with no more than one arithmetical error.
	A1	For the correct inequality.
(b)	M1	For attempting to solve the QE to find **two** critical values.
	For the dM and A marks allow any acceptable notation. For example; $-\frac{5}{3} \leqslant x \cap x \leqslant 2$ or, $-\frac{5}{3} \leqslant x$ **and** $x \leqslant 2$ The region must however indicate an INSIDE region	
	dM1	Forms an inside region with **their** cv's This mark is dependent on the first M mark in (b)
	A1	For the correct region with the correct inequalities.
(c)	B1ft	For the correct combined inside region. Penalise the incorrect inequality from part (b) $<$ in place of $\leqslant$ only once. Even if the **correct** inequality does not follow from their work, award this mark.

Type MS · Quadratic inequalities

Answer 8 of 10 · Question 2 part b · 3 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Quadratic inequalities

Answer 9 of 10 · Question 2 part b · 4 marks

2020 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

2(a)	$x \leqslant -1$	B1	(1)
(b)	$8x^2 + 10x - 3 (< 0)$		
	$(4x - 1)(2x + 3) (< 0)$	M1	
	$x = \frac{1}{4} \quad x = -\frac{3}{2}$	A1A1	
	$-\frac{3}{2} < x < \frac{1}{4}$	A1ft	(4)
(c)	$-\frac{3}{2} < x \leqslant -1$	B1	(1)
			[6]

(a)	
B1	For $x \leqslant -1$
(b)	Accept decimals in (b) and (c)
NB	The first 3 marks are for finding the critical values. Allow with < or = used.
M1	Attempt to obtain the critical values by solving their 3TQ by any valid method.
A1	Either CV correct
A1	Second CV correct. Award these 2 marks if correct CVs seen in an inequality.
A1ft	Inequality formed to indicate the values between their CVs. Must use < (Can be written in set language).
NB	If CVs incorrect and only shown in the inequality, award 0/4 if no working shown for solving their 3TQ: if working shown M1A0A0A1 is available.
(c)	
B1	For $-\frac{3}{2} < x \leqslant -1$ (no ft)

Type MS · Quadratic inequalities

Answer 9 of 10 · Question 2 part b · 4 marks

2020 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Quadratic inequalities

Answer 10 of 10 · Question 2 part b · 2 marks

2025 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

2(a)	$3-2x=3x^2+9x-17 \Rightarrow 3x^2+11x-20[=0]$ oe eg $x^2+\frac{11}{3}x-\frac{20}{3}[=0]$ eg $(3x-4)(x+5)[=0] \Rightarrow x=\frac{4}{3}$ oe, -5	M1 M1A1 [3]
(b)	$[3-2x \leqslant 3x^2+9x-17 \Rightarrow 3x^2+11x-20 \leqslant 0]$ $x \leqslant$ "-5", $x \geqslant$ "$\frac{4}{3}$" $x \leqslant -5$, $x \geqslant \frac{4}{3}$ oe	M1 A1 [2]
		Total 5 marks

Type MS · Quadratic inequalities

Answer 10 of 10 · Question 2 part b · 2 marks

2025 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	Sets the line = curve and forms the correct 3TQ The correct 3TQ alone would imply this mark. We will condone a missing = 0
	M1	Solves their 3TQ by any minimally acceptable method (see general guidance) Correct values alone for x will **not** imply this and the previous method mark in this question, as the demand of the question, tells the candidates to 'use algebra' We will condone a missing = 0 Not a fully dependent method mark, but is dependent on correctly setting line = curve.
	A1	For both $x=\frac{4}{3}$ **and** -5 Accept for $\frac{4}{3}$, $1.\dot{3}$ or 1.33.... or 1.3^{r} or 1.33 (or better) or any equivalent fraction. We need to see as a minimum, **both** the correct 3TQ written down **and** a correct method to solve this equation. As a minimum: Factorising - we must see $(3x-4)(x+5)$ or $(3x-4)$ **and** $(x+5)$ separately or **any** correct factorisation. We will condone, for example, $3x^2+11x-20$ followed by $\left(x-\frac{4}{3}\right)(x+5)$ Using the formula, we need a correct substitution into the formula. This doesn't need to be simplified, but allow one such as $\frac{-11\pm\sqrt{121+240}}{6}$ or $\frac{-11\pm\sqrt{361}}{6}$ as minimum. Completing the square, need to see $3\left(x+\frac{11}{6}\right)^2-\frac{361}{12}$ or $3\left(x+\frac{11}{6}\right)^2=\frac{361}{12}$ or $\left(x+\frac{11}{6}\right)^2=\frac{361}{36}$
(b)	M1	Chooses the outside region, for their values of x which if incorrect, must come clearly from part (a). It is possible to attain M1 A1 if the (ful)l working isn't shown in (a). For a re-start, possibly using a calculator, finding the correct values and choosing the region correctly. possible to be awarded M1 A1 for this approach also. For this mark allow < for $\leqslant$ and > for $\geqslant$ Allow $x\leqslant -5,\quad x\geqslant 1.3$ or $x<-5,\quad x>1.3$ Allow eg $\frac{4}{3}\leqslant x\leqslant -5$
	A1	For the correct region specified correctly. We do not need to see the word 'and'. Accept for $\frac{4}{3}$, $1.\dot{3}$ or 1.33.... or 1.3^{r} or 1.33 (or better) Accept equivalences eg $(-\infty,-5]$ or $[-\infty,-5]$ and $\left[\frac{4}{3},\infty\right]$ or $\left[\frac{4}{3},\infty\right)$ For this mark do not allow < for $\leqslant$ and > for $\geqslant$ **Do not allow** $x\leqslant -5,\quad x\geqslant 1.3$ or $\frac{4}{3}\leqslant x\leqslant -5$

Type MS · Quadratic inequalities

Answer 10 of 10 · Question 2 part b · 2 marks

2025 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks